Report of assignment-2

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Task 1:

We generated 1,200 observations of French maize plants: 900 healthy (y=0, 75%) and 300 unhealthy due to pest attacks (y=1, 25%). The dataset contains 5 features: 4 numerical and 1 categorical.

Column	Meaning	Generation
`f1`	Ear length, cm	Healthy: N(20, 4); unhealthy: N(15, 4). Both standard deviations are 2 cm.
`f2`	Standardized leaf reflectance anomaly	Healthy: N(0, 0.49). Unhealthy: equal mixture of N(-3, 0.49) and N(3, 0.49).
`f3`	Standardized leaf pigment deviation	N(0, 1) in both classes, jointly generated with f4.
`f4`	Standardized leaf fluorescence deviation	N(0, 1) in both classes; correlation with f3 is -0.8 for healthy and +0.8 for unhealthy plants.
`cat`	Pest-management treatment	Four levels with exact class-dependent counts shown below.

Category counts per class

Treatment	Healthy	Unhealthy	Healthy %	Unhealthy %
No treatment	180	150	20.0	50.0
Biological control	225	90	25.0	30.0
Standard treatment	225	57	25.0	19.0
Enhanced protection	270	3	30.0	1.0

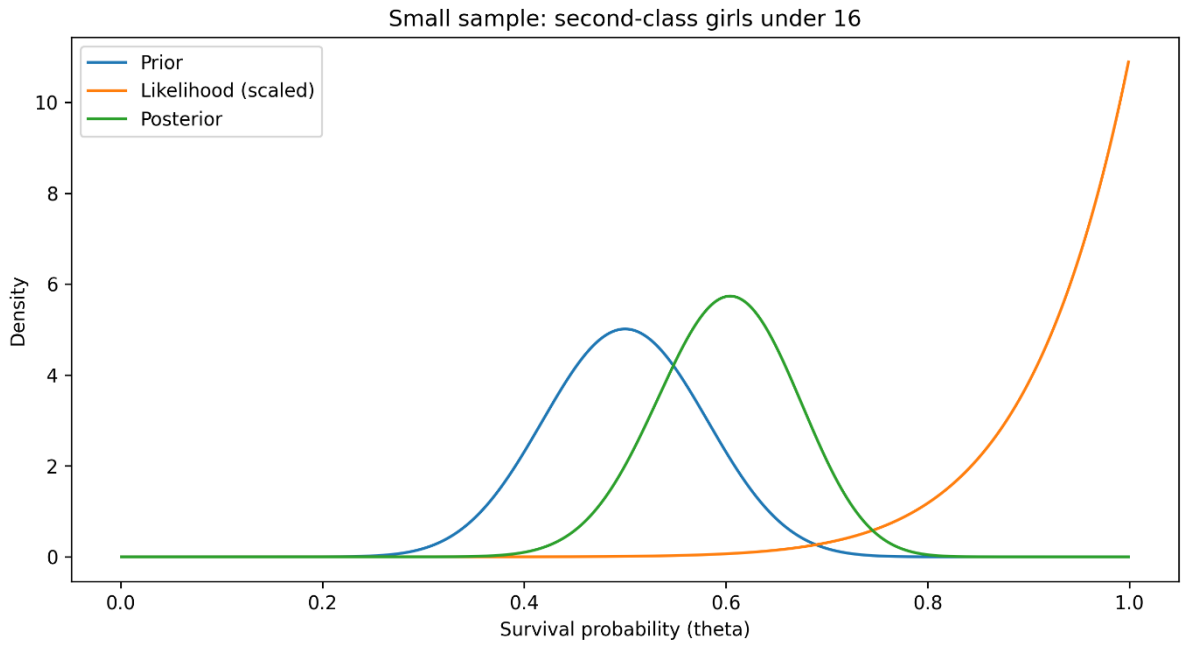
Task 2a:

The Beta-Binomial model estimates the probability of an unhealthy plant using a prior and observed data to obtain a posterior distribution. Unhealthy plants are defined as successes. The observed unhealthy rate is 300/1,200 = 25%, while the posterior mean is approximately 25.08%. These values are close because the large sample makes the likelihood more influential than the prior.

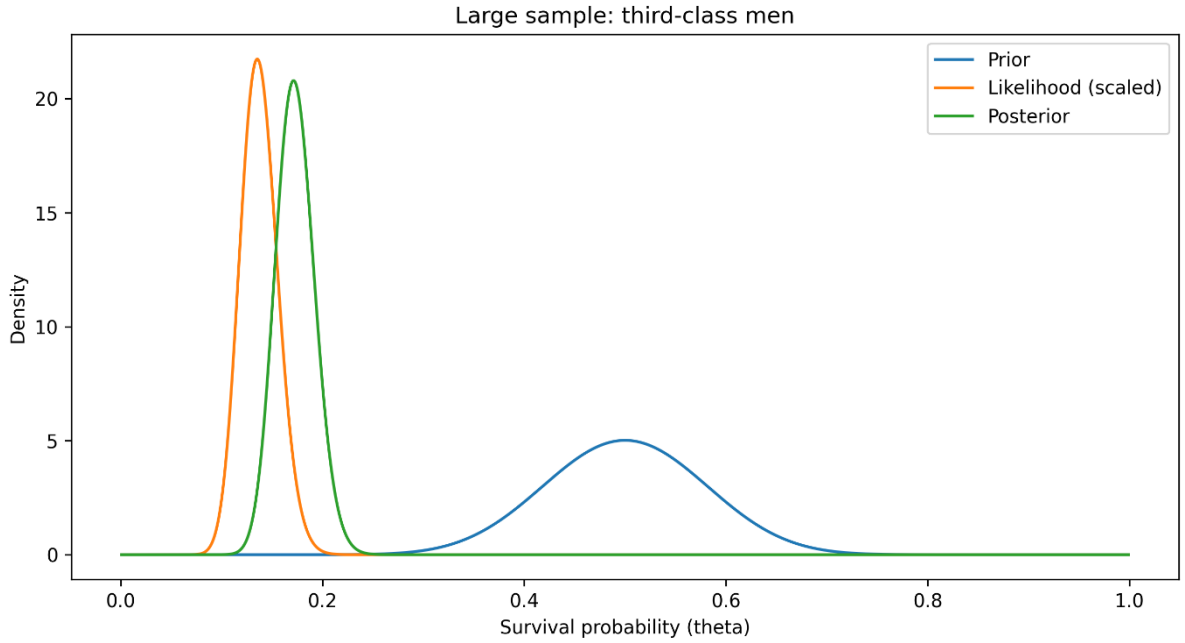
The 95% credible interval is 22.68%–27.57%, meaning we are 95% confident that unhealthy rate lies between this range.

The posterior mode is the value at the peak of the posterior density.

Task 2b:



For second-class girls under 16, all 10 survived, yet the probability to survive is low, 60%. The Beta(20,20) prior pulls the estimate towards 50%, demonstrating its strong influence in small samples.



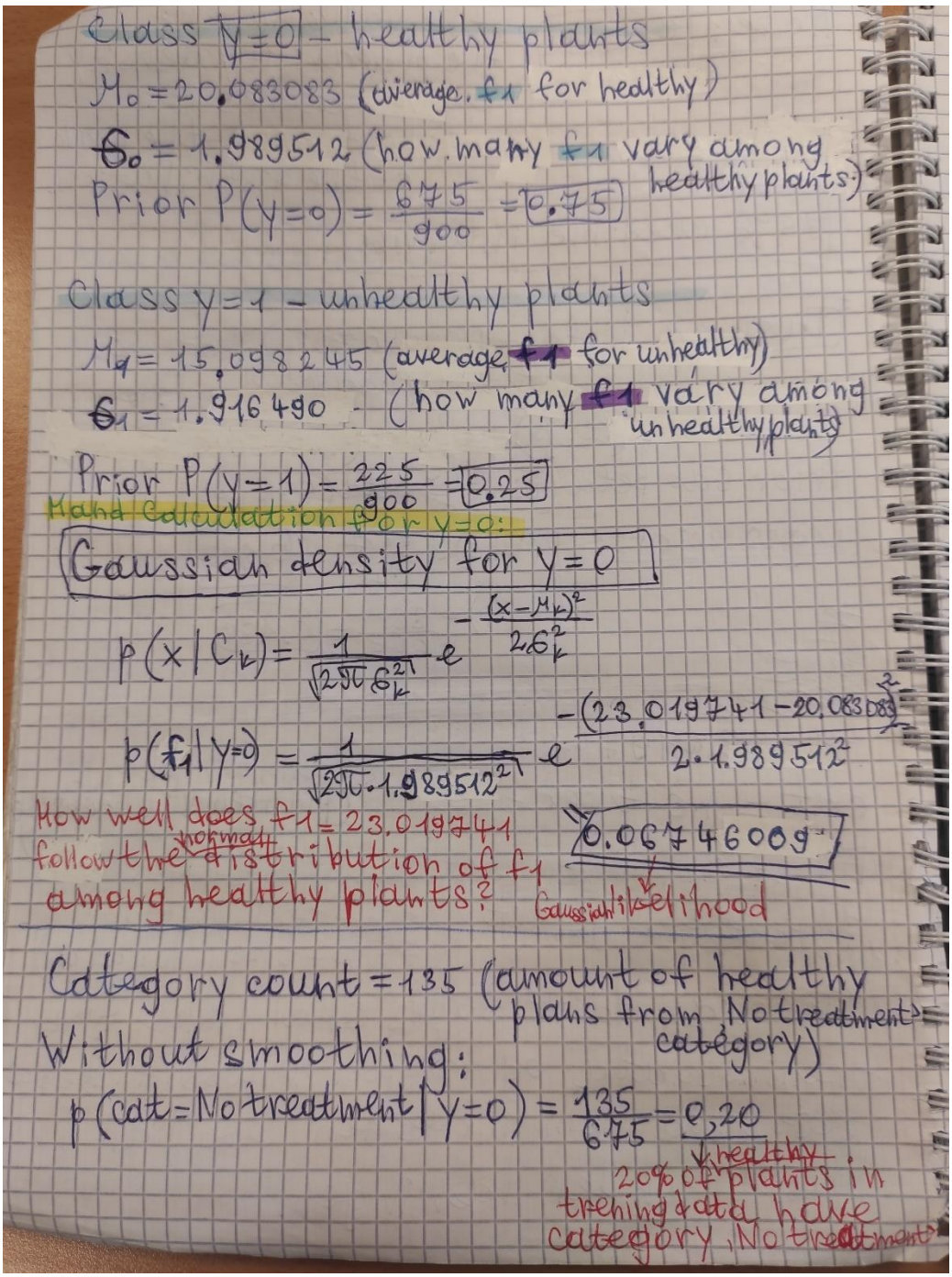
For third-class men, the larger sample makes the data more influential. Its narrower credible interval indicates lower posterior uncertainty.

Task 3a (Hand calculation):

We chose Gaussian Naïve Bayes because it is simple and fast. It assumes numerical features follow Gaussian distributions within each class.

Recipe for hand calculation:

1) Split training data by class
2) Compute *mean* and *std. deviation* for every feature, per class
3) Compute the class priors
4) For a new point, evaluate every density (likelihoods)
5) Multiply: prior * all likelihoods
6) Pick the largest; divide by the total to get a probability (normalization)



But the code uses Laplace smoothing, in order to avoid that a categorical likelihood becomes zero, if category doesn't appear in trening data for a specific class.

With Laplace smoothing:

$\frac{135+1}{675+4} = 0.20029455$ → to avoid zero possibility

4 → amount of categories

categorical likelihood that a plant from No treatment is healthy

So, prior x all likelihoods for y=0 is:

$0.75 \times 0.06746009 \times 0.20029455$

$= 0.010133916 3$ → unnormalized score for y=0

Hand calculation for y=1:

Gaussian density for y=1

$$p(f_1 \mid y=1) = \frac{1}{\sqrt{2\pi \cdot 1.916490^2}} \cdot e^{\frac{-(23.019741-15.098245)^2}{2 \cdot 1.916490^2}}$$

$= 0.00004060$

Category count: 110 (amount of unhealthy plants from 'No treatment' category)

Without smoothing:

$\frac{110}{225} = 0.4889$

With Laplace smoothing:

$\frac{110+1}{225+4} = 0.484471616$

So, [prior x all likelihoods] for y=1 is

$0.25 \times 0.000040060 \times 0.48471616$

$= 0.00000049203$ → unnormalized score for y=1

NORMALIZATION

↓

Pick the largest unnormalized score, and do this formula:

$$P(C_k|x) = \frac{P(C_k)\,p(x|C_k)}{\sum_j P(C_j)\,p(x|C_j)}$$

So, the probability that test plant is healthy

$$P(y=0|x) = \frac{0.0101339163}{0.0101388366} \approx 0.99951471$$

winner!

While for unhealthy plant

$$P(y=1|x) = \frac{0.00000049203}{0.0101388366} \approx 0.00048529$$

So, since there is almost 100% chance to get a healthy plant, the model predicts that.

Task 3b:

Result	No smoothing (`alpha = 0`)	Laplace smoothing (`alpha = 1`)
`P(Enhanced protection | y = 0)`	0.303704	0.303387
`P(Enhanced protection | y = 1)`	0.000000	0.004425
`P(y = 1 | x)`	0.000000	0.003235

Laplace smoothing prevents zero categorical probabilities by adding pseudocounts. Without smoothing, “Enhanced protection” had zero observations in class y=1, making its likelihood zero. But smoothing introduces bias, especially in small samples.

Task3c:

Features	Number of features	Mean F1-score	Standard deviation
`f1 + f2 + cat`	3	0.963062	0.012043
`f1 + f2 + f3 + f4 + cat`	5	0.960913	0.009437
`f2`	1	0.947221	0.020014
`f1 + cat`	2	0.859755	0.036231
`f1 + f3 + f4 + cat`	4	0.859616	0.038830
`f1`	1	0.850020	0.033754
`f3 + f4`	2	0.000000	0.000000

Model	Accuracy	Precision	Recall	F1-score
All five features	0.973333	0.924051	0.973333	0.948052
Selected subset (`f1 + f2 + cat`)	0.976667	0.935897	0.973333	0.954248

The feature subset f1+f2+cat performed best, with a mean cross-validation F1-score of 0.963062. It outperformed larger subsets despite using fewer features, showing that additional features don’t improve classification.

Task 4:

Split	Seed	Model	Train n	Train y = 1 n (%)	Test n	Test y = 1 n (%)	Test error (%)	Sensitivity (%)	Status / note
A: Biased	-	Naive Bayes	300	300 (100.00%)	900	0 (0.00%)	N/A	N/A	[1]
A: Biased	-	Decision Tree	300	300 (100.00%)	900	0 (0.00%)	100.00	N/A	[2]
B: Random	42	Naive Bayes	960	237 (24.69%)	240	63 (26.25%)	3.33	95.24	OK
B: Random	42	Decision Tree	960	237 (24.69%)	240	63 (26.25%)	4.17	92.06	OK
B: Random	43	Naive Bayes	960	232 (24.17%)	240	68 (28.33%)	1.25	97.06	OK
B: Random	43	Decision Tree	960	232 (24.17%)	240	68 (28.33%)	1.67	97.06	OK
B: Random	44	Naive Bayes	960	241 (25.10%)	240	59 (24.58%)	2.08	96.61	OK
B: Random	44	Decision Tree	960	241 (25.10%)	240	59 (24.58%)	2.50	96.61	OK
B: Random	45	Naive Bayes	960	239 (24.90%)	240	61 (25.42%)	2.50	100.00	OK
B: Random	45	Decision Tree	960	239 (24.90%)	240	61 (25.42%)	4.58	93.44	OK
B: Random	46	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	0.42	100.00	OK
B: Random	46	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	0.83	96.67	OK
C: Stratified	42	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	2.92	96.67	OK
C: Stratified	42	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	4.17	96.67	OK
C: Stratified	43	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	1.67	98.33	OK
C: Stratified	43	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	1.67	96.67	OK
C: Stratified	44	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	2.92	98.33	OK
C: Stratified	44	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	2.08	96.67	OK
C: Stratified	45	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	2.50	100.00	OK
C: Stratified	45	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	1.67	98.33	OK
C: Stratified	46	Naive Bayes	960	240 (25.00%)	240	60 (25.00%)	1.25	100.00	OK
C: Stratified	46	Decision Tree	960	240 (25.00%)	240	60 (25.00%)	1.25	100.00	OK

1) In split A, all 300 minority-class observations (y=1) were assigned to training and all 900 majority-class observations (y=0) to testing. Thus, training contained 100% y=1, while testing contained 0%,so sensitivity was undefined.

2) Sensitivity varied less across seeds in split C than in split B. The standard deviation went from 2.13 to 1.39 percentage points for Naive Bayes and from 2.26 to 1.49 percentage points for the Decision Tree.

3) Error rate alone is misleading due to imbalanced dataset: always predicting y=0 gives 75% accuracy without detecting any unhealthy plants.

4) In split C, the Decision Tree had a slightly lower mean error rate than Naïve Bayes (2.17% vs 2.25%), while Naïve Bayes had higher mean sensitivity (98.67% vs 97.67%). Thus, both models performed similarly, but Naïve Bayes detected unhealthy plants slightly better.